Skeletal Dysplasia, Mild, with Joint Laxity and Advanced Bone Age (SDJLABA / CSGALNACT1-CDG)

Disease: Skeletal Dysplasia Mild With Joint Laxity And Advanced Bone Age (SDJLABA) **MONDO ID:** MONDO:0030029 · **OMIM:** 618870 · **Causal gene: CSGALNACT1** (HGNC:24290; gene OMIM 616615) **Category:** Mendelian, autosomal recessive

Summary

Skeletal Dysplasia, Mild, with Joint Laxity and Advanced Bone Age (SDJLABA) — also termed **CSGALNACT1-CDG** (a congenital disorder of glycosylation) — is an ultra-rare autosomal-recessive osteochondrodysplasia caused by **biallelic loss-of-function or hypomorphic variants in CSGALNACT1**. This gene encodes **chondroitin sulfate N-acetylgalactosaminyltransferase-1 (CSGalNAcT-1, also called ChGn-1; UniProt Q8TDX6)**, the Golgi enzyme that *initiates* chondroitin/dermatan sulfate (CS/DS) glycosaminoglycan (GAG) chains by transferring the first N-acetylgalactosamine (GalNAc) residue onto the GAG–protein linker tetrasaccharide (GlcA-Gal-Gal-Xyl-O-Ser) attached to proteoglycan core proteins such as aggrecan. The disease belongs to the broader group of genetic skeletal disorders caused by defective GAG biosynthesis, mechanistically adjacent to Desbuquois dysplasia and the linkeropathies.

The core clinical picture across the ~5 reported patients comprises **short stature (often prenatal-onset and disproportionate), generalized joint laxity/hypermobility, advanced bone age, mild facial dysmorphism, axial malalignment of the lower limbs/feet, and occasional mild language delay**. Advanced bone age is a recurrent distinguishing feature. The disorder is **mild and non-progressive**, with normal survival, normal-to-near-normal cognition, and no reported major cardiac or renal involvement. Reduced CSGalNAcT-1 activity lowers chondroitin sulfate on aggrecan and other proteoglycans (~50% reduction in knockout-mouse cartilage), disorganizing the growth-plate extracellular matrix and impairing endochondral ossification. Patient fibroblasts display altered chondroitin, dermatan, and heparan sulfate levels.

Management is entirely **supportive** — orthopedic surveillance and correction of limb malalignment/scoliosis, physiotherapy for joint laxity, speech/language therapy when indicated, and genetic counseling — as there is no disease-specific, pharmacological, gene, cell, or RNA therapy, and no registered clinical trials. The **Csgalnact1-knockout mouse** faithfully recapitulates the skeletal phenotype (mild dwarfism, ~50% reduced cartilage CS, disorganized growth-plate chondrocyte columns, abnormal type-II collagen organization), providing a validated model. gnomAD constraint metrics (pLI ≈ 0; observed/expected LoF ≈ 1.02; missense Z = −4.19) confirm that CSGALNACT1 is loss-of-function- and missense-tolerant at the heterozygous level, consistent with a recessive mechanism without haploinsufficiency.

Section 1 — Disease Information

Overview. SDJLABA is a Mendelian, autosomal-recessive, mild non-progressive skeletal dysplasia. It arises from biallelic defects in **CSGALNACT1**, the enzyme that initiates chondroitin sulfate chain assembly on proteoglycans. Because the defect lies in a glycosylation pathway, the disorder is also classified as a **congenital disorder of glycosylation (CSGALNACT1-CDG)**. It is characterized clinically by short stature, joint laxity, advanced bone age, mild facial dysmorphism, and limb malalignment.

Key identifiers.

Resource	Identifier
OMIM (disease)	618870
MONDO	MONDO:0030029
Gene (symbol / HGNC)	CSGALNACT1 / HGNC:24290
Gene OMIM	616615
Ensembl	ENSG00000147408
UniProt (protein)	Q8TDX6
Locus	8p21.3 (chr8:19,404,161–19,758,029, GRCh38)
Orphanet / ICD-10 / ICD-11 / MeSH	No dedicated code identified (ultra-rare)

Synonyms / alternative names. SDJLABA; CSGALNACT1-CDG; CSGalNAcT-1 deficiency; chondroitin sulfate N-acetylgalactosaminyltransferase-1 deficiency skeletal dysplasia.

Source of information. All information is derived from **aggregated disease-level resources** (OMIM, gnomAD) and from **individual-patient primary case reports** (three primary clinical papers describing ~5 patients) plus mechanistic studies in knockout mice and in vitro systems. There is no EHR- or registry-scale dataset for this ultra-rare disease.

Section 2 — Etiology

Primary cause. The disease is **purely genetic**. It is caused by **biallelic (compound heterozygous or homozygous) loss-of-function / hypomorphic variants in CSGALNACT1**. No environmental, infectious, or acquired cause is implicated.

Three independent reports establish causality in ≥5 patients: - **Vodopiutz et al. 2017** (PMID: 27599773): compound heterozygous large intragenic deletion + missense p.Pro384Arg, Desbuquois-like phenotype — *"This is the first description of bi-allelic loss-of-function mutations in CSGALNACT1 that produce a skeletal dysplasia."* - **Meyer et al. 2019** (PMID: 31325655): homozygous frameshift variant in a 12-year-old boy — *"Exome sequencing revealed a biallelic loss of function mutation in CSGALNACT1, which encodes chondroitin sulfate N-acetylgalactosaminyltransferase 1 and plays a major role in the chondroitin sulfate chain biosynthesis."* - **Mizumoto et al. 2020** (PMID: 31705726): two unrelated patients — compound heterozygous c.1294G>T (p.Asp432Tyr) + exon 4 deletion (removing the start codon), and homozygous c.791A>G (p.Asn264Ser) — concluding *"biallelic loss-of-function mutations in CSGALNACT1 disturb glycosaminoglycan synthesis and cause a mild skeletal dysplasia with advanced bone age, CSGALNACT1-CDG."*

Genetic risk factors. The causal variants themselves are the only risk factor. Consanguinity / shared founder haplotypes raise risk for homozygous genotypes (as inferred from the homozygous cases). No susceptibility loci or modifier genes have been identified.

Environmental risk / protective factors. None known or expected for a monogenic recessive disorder. No lifestyle, toxin, occupational, or dietary factor has been implicated.

Gene–environment interactions. None documented.

Section 3 — Phenotypes

The phenotype is a **mild, generally non-progressive skeletal dysplasia** with connective-tissue features. HPO term suggestions are provided per feature.

Phenotype	Type	HPO suggestion	Onset	Severity	Frequency (of ~5 cases)
Short stature (often disproportionate, prenatal-onset possible)	Physical / clinical sign	HP:0004322 (Short stature)	Prenatal–childhood	Mild–moderate	Most/all
Generalized joint laxity / hypermobility	Clinical sign	HP:0001388 (Joint laxity) / HP:0001382 (Joint hypermobility)	Childhood	Mild–moderate	Most/all
Advanced bone age	Radiographic / imaging finding	HP:0003504 (Advanced skeletal maturation)	Childhood	Characteristic	Recurrent (distinguishing)
Facial dysmorphism	Physical manifestation	HP:0001999 (Abnormal facial shape)	Congenital/ childhood	Mild	Multiple
Axial malalignment of lower limbs/feet	Clinical sign	HP:0002979 (Bowing of limbs) / HP:0040069 (Abnormal lower limb bone morphology)	Childhood	Mild	Reported
Mild language / speech delay	Behavioral / developmental	HP:0000750 (Delayed speech and language development)	Childhood	Mild	Subset (Mizumoto cases)

Characteristics summary. Onset ranges from prenatal (severe end — Vodopiutz index infant with *"prenatal-onset disproportionate short stature, joint laxity, and radiographic findings typical for"* Desbuquois dysplasia) to childhood recognition. Severity is **mild** overall;

progression is **stable / non-progressive** (Meyer et al.: *"mild and non-progressive skeletal dysplasia with joint laxity"*). Cognition is normal or near-normal, with at most mild language delay — not intellectual disability (Mizumoto et al. described *"short stature with advanced bone age, facial dysmorphism, and mild language delay"*).

Quality-of-life impact. No formal QoL instrument (EQ-5D/SF-36/PROMIS) data exist. Given mild, non-progressive disease with normal lifespan, QoL impact is expected to be modest, driven mainly by orthopedic concerns (limb malalignment, joint instability) and, in a subset, speech/language support needs.

Section 4 — Genetic / Molecular Information

Causal gene. CSGALNACT1 (8p21.3; ENSG00000147408; HGNC:24290; gene OMIM 616615). Protein: CSGalNAcT-1 / ChGn-1 (UniProt Q8TDX6), a Golgi glycosyltransferase.

Reported pathogenic / likely-pathogenic variants.

Patient (study)	Genotype	Variant type	Functional consequence
Vodopiutz 2017 (PMID: 27599773)	Large intragenic deletion / p.Pro384Arg	Structural (deletion) + missense	Reduced GalNAc-transferase activity (Arg-384 mutant)
Meyer 2019 (PMID: 31325655)	Homozygous frameshift	Frameshift (LoF)	Loss of function
Mizumoto 2020 — patient 1 (PMID: 31705726)	c.1294G>T (p.Asp432Tyr) / exon 4 deletion	Missense + deletion removing start codon	Reduced enzyme activity (missense); null (start-codon loss)
Mizumoto 2020 — patient 2 (PMID: 31705726)	Homozygous c.791A>G (p.Asn264Ser)	Missense	Significantly reduced CSGalNAcT-1 activity

Variant classification. Pathogenic / likely pathogenic by ACMG criteria (biallelic LoF, functional enzyme assays demonstrating reduced activity, segregation, rarity in gnomAD). Missense variants were validated biochemically to reduce GalNAc-transferase activity.

Functional consequence. Loss of function (recessive). No gain-of-function or dominant-negative mechanism; heterozygous carriers are unaffected.

Allele frequency / population constraint. gnomAD (GRCh38): pLI = 1.6×10^{-18} (≈0); observed/expected LoF = 1.02 (90% CI 0.82–1.29) — **no depletion of LoF variants**; missense Z = −4.19 (**missense-tolerant**). These metrics indicate CSGALNACT1 tolerates heterozygous loss of function, consistent with a recessive disease mechanism and no haploinsufficiency.

Modifier genes / epigenetics / chromosomal abnormalities. No modifier genes, disease-specific epigenetic changes, or large-scale chromosomal abnormalities are implicated. The exon-4 deletion and the large intragenic deletion are intragenic structural variants, not cytogenetic rearrangements.

Section 5 — Environmental Information

No **environmental**, **lifestyle**, or **infectious** factors contribute to this monogenic recessive disorder. There are no relevant toxin, radiation, dietary, or pathogen associations. This section is **not applicable** beyond noting the purely genetic etiology.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic LoF/hypomorphic variants in CSGALNACT1** reduce/abolish functional CSGalNAcT-1 (ChGn-1) enzyme — *demonstrated* by enzyme assays on missense mutants.
2. Reduced CSGalNAcT-1 activity **leads to** deficient transfer of the first (and elongating) GalNAc residue onto the GAG–protein linker tetrasaccharide (GlcA-Gal-Gal-Xyl-O-Ser) — *demonstrated* enzymology; this is the **chondroitin sulfate chain-initiation** step (*"CSGalNAcT-1 … initiates chondroitin sulfate (CS) chain biosynthesis on the so-called GAG-protein linker region tetrasaccharide"*, Vodopiutz 2017).
3. Impaired CS chain initiation **results in** reduced chondroitin/dermatan sulfate GAG on proteoglycan core proteins (notably **aggrecan**); patient fibroblasts show *"altered levels of chondroitin, dermatan, and heparan sulfate moieties … compared to controls"* (branch: linker/precursor competition can also perturb heparan sulfate) — *demonstrated* in patient cells and model cartilage (~50% CS reduction in mouse).

4. Reduced CS on aggrecan and other proteoglycans **leads to** a disorganized growth-plate extracellular matrix: shorter, disordered proliferative-zone chondrocyte columns, reduced ECM volume, and abnormally aggregated/disarranged type-II collagen fibrils — *demonstrated* histologically in knockout mice.
5. Growth-plate ECM disorganization **impairs endochondral ossification and chondrocyte column organization** — *demonstrated* in mouse; **results in** short stature and skeletal dysplasia.
6. Altered proteoglycan ECM at joints and connective tissue **results in** joint laxity/hypermobility and limb malalignment (*inferred* from ECM/collagen disorganization).
7. Advanced bone age arises from dysregulated chondrocyte maturation/ossification timing (*inferred*; mechanism not fully demonstrated).
8. **Branch — neurodevelopment:** CSGalNAcT-1-dependent CS is concentrated in **perineuronal nets (PNNs)** that regulate synaptic plasticity and critical-period timing; T1-knockout mice *"show extensive axon regeneration following spinal cord injury, as well as the loss of onset of ocular dominance plasticity."* Reduced CS **may lead to** altered neuronal plasticity, providing a plausible basis for the **mild language delay** seen in a subset (*inferred*; only mild language delay — not intellectual disability — is observed clinically).

Supporting detail

- **Molecular pathways.** GAG / proteoglycan biosynthesis (chondroitin/dermatan sulfate assembly on the linker tetrasaccharide). KEGG: glycosaminoglycan biosynthesis — chondroitin/dermatan sulfate (hsa00532). Enzyme: EC 2.4.1.174 (chondroitin sulfate GalNAc-transferase).
- **Cellular processes.** Chondrocyte proliferation/organization in the growth plate; ECM assembly; aggrecan metabolism (decreased aggrecan and link protein 1, increased processed aggrecan in KO cartilage).
- **Protein dysfunction.** Loss of catalytic activity of a Golgi type-II membrane glycosyltransferase; reduced GalNAc transfer to the linker. Missense substitutions (p.Pro384Arg, p.Asp432Tyr, p.Asn264Ser) lower activity; frameshift/start-codon-loss alleles are null.
- **Biochemical abnormality.** CS chain-initiation deficiency → ~50% reduction of cartilage CS (chain length preserved → defect is in *initiation*, not elongation) in the mouse model; altered CS/DS/HS in patient fibroblasts.
- **Tissue damage mechanism.** Not injury/inflammation-driven; rather a developmental ECM-assembly deficit affecting cartilage/connective tissue.

- **Immune/metabolic involvement.** Not central to disease. (Of note, CSGalNAcT1-KO mice show *milder* experimental autoimmune encephalomyelitis, reflecting CS's role in neuroimmune responses — relevant to CS biology but not to patient pathology.)

Ontology suggestions

- **GO biological process:** GO:0030206 (chondroitin sulfate biosynthetic process); GO:0030204 (chondroitin sulfate metabolic process); GO:0001958 (endochondral ossification); GO:0030198 (extracellular matrix organization); GO:0051216 (cartilage development).
- **GO cellular component:** GO:0000139 (Golgi membrane); GO:0005794 (Golgi apparatus); GO:0062023 (collagen-containing extracellular matrix).
- **GO molecular function:** GO:0047238 (CS GalNAc-transferase activity).
- **CL cell types:** CL:0000138 (chondrocyte); CL:0000057 (fibroblast); CL:0000540 (neuron, for PNN branch).
- **CHEBI:** CHEBI:37397 (chondroitin sulfate); CHEBI:52396 (dermatan sulfate); CHEBI:28815 (N-acetylgalactosamine).

Section 7 — Anatomical Structures Affected

Organ / system level. Primary involvement is the **skeletal system** — growth plates, long bones, and joints. **Connective tissue** is affected (joint laxity). A **secondary, mild nervous-system** contribution is plausible (speech/language) via CS in perineuronal nets. No cardiac, renal, hepatic, or hematologic involvement is reported.

Tissue / cell level. Affected tissues: **cartilage** (hyaline growth-plate cartilage) and connective tissue; affected cells: **chondrocytes** (CL:0000138), particularly proliferative-zone chondrocytes, and **fibroblasts** (CL:0000057). Neurons bearing perineuronal nets (CL:0000540) are implicated in the inferred neurodevelopmental branch.

Subcellular level. The enzymatic defect localizes to the **Golgi apparatus / Golgi membrane** (GO:0005794 / GO:0000139), where GAG chains are assembled; the downstream deficit manifests in the **extracellular matrix** (GO:0062023).

Localization / lateralization. Skeletal involvement is **generalized and bilateral/symmetric** (systemic dysplasia). UBERON terms: UBERON:0002544 (cartilage), UBERON:0004765 (skeletal element), UBERON:0002481 (bone tissue), UBERON:0001474 (bone element), UBERON:0000982 (skeletal joint), UBERON:0004624 (growth plate).

Section 8 — Temporal Development

- **Onset.** Congenital to pediatric. Short stature can be **prenatal-onset** (severe end, Vodopiutz index infant); most features are recognized in **childhood**. Onset pattern is **insidious/chronic** (developmental), not acute.
- **Progression.** The disorder is **mild and non-progressive** (Meyer et al.). No defined staging; disease course is **stable/chronic-lifelong** without relapsing-remitting or episodic patterns.
- **Patterns.** No remission concept applies (structural/developmental). **Critical periods** are developmental: growth-plate activity during childhood/puberty is the window where orthopedic complications (limb malalignment, possible scoliosis) emerge and where monitoring/intervention is most relevant.

Section 9 — Inheritance and Population

- **Epidemiology. Ultra-rare**: only ~5 patients from 3 primary reports described as of 2026. No prevalence/incidence estimates exist; too rare for registry quantification (likely <1 per million).
- **Inheritance. Autosomal recessive** — biallelic variants required. Confirmed by homozygous and compound-heterozygous genotypes and by gnomAD showing no LoF constraint (pLI ≈ 0), i.e., heterozygous carriers are unaffected. Meyer 2019 identified *"a second case and the first juvenile patient with a homozygous frameshift variant in CSGALNACT1"*; Mizumoto 2020 *"investigated two unrelated patients."*
- **Penetrance / expressivity.** Presumed high penetrance in biallelic carriers; **variable expressivity** across the reported spectrum (prenatal-onset Desbuquois-like at the severe end to milder childhood presentations).
- **Genetic anticipation.** Not applicable (not a repeat-expansion disorder).

- **Germline mosaicism / founder effects.** Not specifically documented; homozygous genotypes suggest **consanguinity or shared founder haplotypes** in some families.
- **Carrier frequency.** Not formally established; expected very low given rarity. gnomAD does not flag CSGALNACT1 LoF depletion, consistent with rare recessive carriers.
- **Demographics.** Both sexes affected (male patient explicitly in Meyer 2019); patients are geographically scattered (unrelated families across reports). No ethnic predilection identified. Age distribution spans infancy to adolescence at reporting.

Section 10 — Diagnostics

- **Genetic testing (definitive).** Diagnosis rests on **molecular identification of biallelic CSGALNACT1 variants**. **Whole-exome sequencing (WES)** is the primary diagnostic modality used across reports (*"Exome sequencing revealed a biallelic loss of function mutation in CSGALNACT1"* — Meyer 2019). WGS or skeletal-dysplasia/GAG-biosynthesis gene panels including CSGALNACT1 are alternatives. CMA/karyotyping may detect the large intragenic/exon deletions but cannot resolve point variants.
- **Biochemical / functional confirmation.** Reduced **CSGalNAcT-1 enzyme activity** assays for missense variants and **altered CS/DS/HS GAG levels in patient fibroblasts** provide functional validation (Mizumoto 2020). These are research-grade confirmatory tests.
- **Imaging. Skeletal radiography** is central — demonstrates **advanced bone age** (HP:0003504), Desbuquois-like features at the severe end, and limb malalignment. Hand/wrist bone-age X-ray is a key recurrent finding.
- **Clinical criteria / differential diagnosis.** No formal diagnostic criteria exist. The differential includes other **GAG-biosynthesis skeletal disorders**: **Desbuquois dysplasia** (CANT1 — nucleotide-sugar availability; mechanistically adjacent), **diastrophic dysplasia** (abnormal GAG sulfation, SLC26A2), and the **linkeropathies** (XYLT1/2, B4GALT7, B3GALT6, B3GAT3). Tsujioka 2026 (PMID: 42094029) notes *"Defective linker formation causes distinctive phenotypes termed linkeropathy."* B3GAT3 linkeropathy (PMID: 31196143) typically presents with more severe/consistent skeletal dysplasia, facial dysmorphology, spatulate distal phalanges, and variable cardiac defects, distinguishing it from the milder CSGALNACT1 phenotype.
- **Screening.** No newborn or population screening exists. **Cascade/carrier testing** of relatives is feasible once a proband's variants are known.

Section 11 — Outcome / Prognosis

- **Survival / mortality. Normal lifespan**; no disease-specific mortality reported. No cardiac/renal/respiratory compromise.
- **Morbidity / function.** Morbidity is **mild**, driven by short stature, joint laxity/instability, limb malalignment, and (in a subset) mild speech/language delay. No severe disability reported.
- **Disease course. Non-progressive** (Meyer 2019: *"mild and non-progressive skeletal dysplasia with joint laxity"*). Complications are orthopedic (limb malalignment, possible scoliosis, joint instability). Cognition is normal-to-near-normal.
- **Quality of life.** No formal instrument data; expected modest impact.
- **Prognostic factors.** Severity appears related to residual enzyme activity (null vs. hypomorphic alleles), though the cohort is too small to quantify genotype–phenotype correlations. No prognostic biomarkers established.

Section 12 — Treatment

There is **no disease-specific, pharmacological, gene, cell, or RNA therapy**, and **no registered clinical trials** (no ClinicalTrials.gov entries). Management is **supportive and symptom-directed**, inferred from the phenotype:

Modality	Intervention	NCIT suggestion
Orthopedic	Surveillance and correction of lower-limb axial malalignment; monitoring/management of possible scoliosis; joint-stability management	NCIT:C15329 (Orthopedic Procedure)
Rehabilitation	Physiotherapy for joint laxity/hypermobility and musculoskeletal strengthening	NCIT:C15415 (Physical Therapy)
Speech/ language	Speech-language therapy when delay present	NCIT:C15190 (Speech Therapy)
Genetics	Genetic counseling (recessive recurrence risk, carrier/ cascade testing)	NCIT:C15764 (Genetic Counseling)
Supportive	General pediatric / skeletal-dysplasia supportive care	NCIT:C15277 (Supportive Care)

Pharmacogenomics / targeted / immunotherapy / advanced therapeutics. Not applicable — none reported or trialed.

Section 13 — Prevention

- **Primary prevention.** Not possible for a monogenic recessive disorder beyond **reproductive counseling**. **Genetic counseling** for at-risk couples (especially consanguineous families or known carriers) and options including **carrier/cascade screening**, **prenatal diagnosis**, or **preimplantation genetic testing (PGT-M)** once familial variants are known.
- **Secondary prevention.** Early orthopedic and developmental assessment to detect and manage limb malalignment, scoliosis, and speech delay.
- **Tertiary prevention.** Orthopedic/physiotherapy management to prevent complications of joint laxity and malalignment.
- **Immunization / public-health / environmental measures.** Not applicable.

Section 14 — Other Species / Natural Disease

- **Taxonomy / orthologs.** Mouse ortholog **Csgalnact1** (NCBI Gene / MGI) is well characterized. Orthologs are conserved across vertebrates, consistent with the enzyme's fundamental role in GAG biosynthesis.
- **Natural disease in other species.** No naturally occurring companion-animal or wildlife disease specifically attributed to CSGALNACT1 was identified in OMIA for this review. The disease is studied primarily through engineered mouse knockouts rather than spontaneous animal disease.
- **Comparative biology.** The mouse knockout reproduces the human skeletal phenotype (see Section 15), demonstrating **evolutionary conservation** of the CS-initiation mechanism in cartilage development.
- **Transmission / zoonosis.** Not applicable (non-infectious genetic disorder).

Section 15 — Model Organisms

Primary model: Csgalnact1-knockout mouse (mammalian), which **recapitulates the human phenotype well**:

Feature (mouse KO)	Finding	Study
Overall	Viable, fertile, **slight dwarfism**	Sato 2011 (PMID: 21148564)
Cartilage CS	**~50% reduction**, chain length unchanged → defect in **initiation**	Sato 2011; Watanabe 2010 (PMID: 20812917)
Growth plate	*"shorter and slightly disorganized chondrocyte columns with a reduced volume of the extracellular matrix principally in the proliferative layer"*	Sato 2011
Aggrecan metabolism	Decreased aggrecan and link protein 1; increased processed aggrecan	Sato 2011
Collagen	*"type-II collagen fibres in developing cartilage were abnormally aggregated and disarranged in the homozygous mutant mice"*	Watanabe 2010
Nervous system	Altered perineuronal nets; axon regeneration after spinal injury; loss of ocular-dominance critical-period plasticity	Igarashi 2018 (PMID: 28987564)
Neuroimmune	Milder experimental autoimmune encephalomyelitis (EAE)	Inada 2021 (PMID: 32839819)

Genetic model types available. Constitutive knockout (characterized). Conditional/humanized models not reported but feasible.

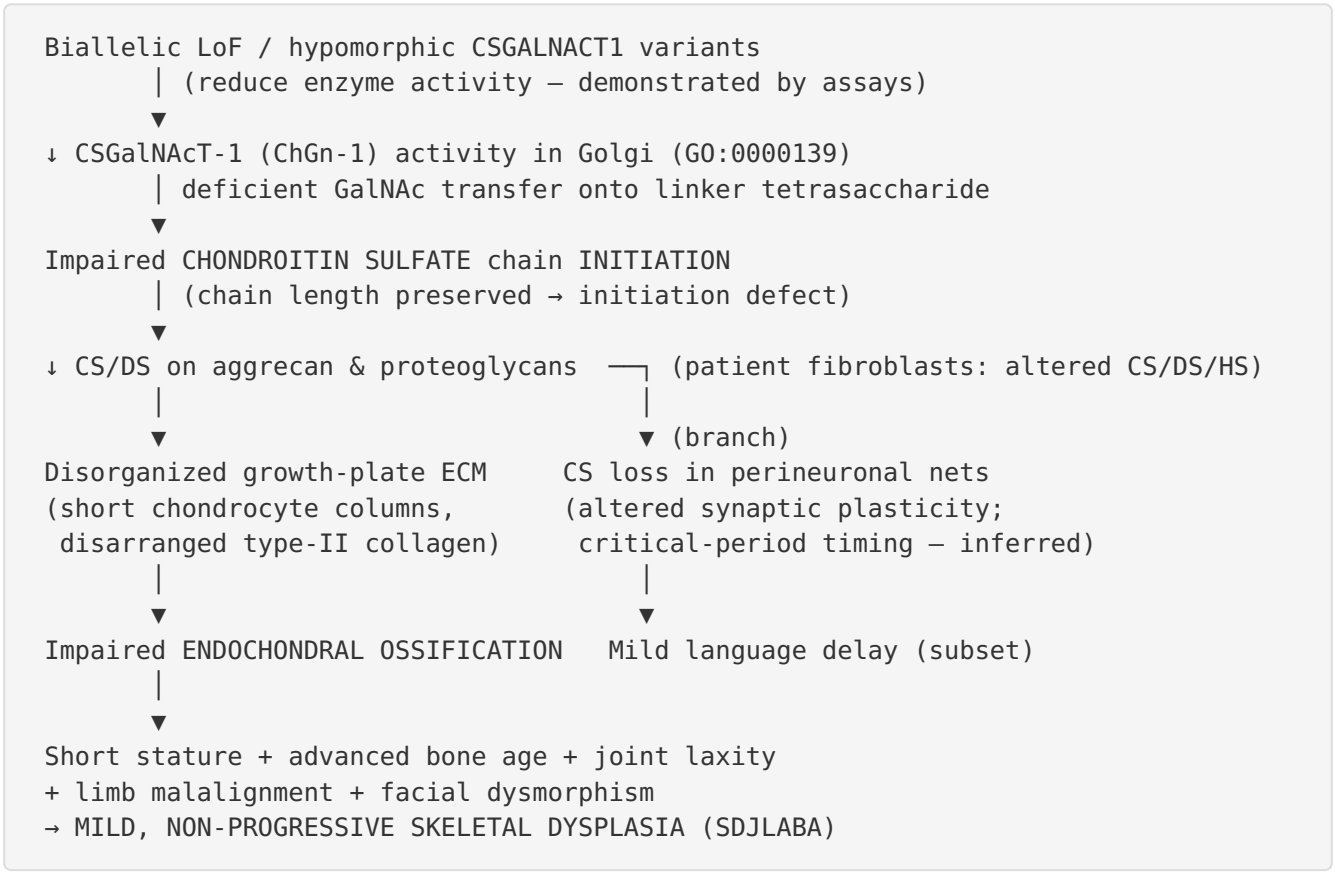
Phenotype recapitulation. Strong for the **skeletal** phenotype (mild dwarfism, reduced CS, growth-plate disorganization). The mouse also enables study of the **neuronal plasticity / PNN** branch relevant to the mild human language-delay feature.

Limitations. The mouse is viable with only slight dwarfism and does not fully model the human "advanced bone age" or facial dysmorphism; cognitive/language phenotypes are not directly translatable. The small human cohort limits genotype–phenotype calibration against the model.

In vitro systems. Patient **fibroblasts** (showing altered CS/DS/HS), recombinant CSGalNAcT-1 enzyme assays, and synthetic linker-oligosaccharide substrate studies (PMID: 32608461) support mechanistic work.

Resources. MGI (Csgalnact1), IMPC/KOMP for engineered alleles.

Mechanistic Model / Interpretation



The disease is best understood as a **cartilage/ECM proteoglycan-assembly deficiency**. The upstream lesion (enzyme loss) and the proximal biochemical consequence (reduced CS initiation) are **demonstrated**; the growth-plate histopathology is **demonstrated in the mouse model**; and the specific links to advanced bone age and to language delay are **plausible but inferred**. The mild, non-progressive nature likely reflects partial redundancy (residual CS synthesis via preserved chain length and overlapping transferases) and the LoF-tolerant constraint profile of the gene.

Evidence Base

PMID	Title (abbrev.)	Role / what it supports
27599773	*CSGalNAcT-1 Deficiency Results in a Mild Skeletal Dysplasia and Joint Laxity*	First patient; establishes biallelic LoF causality; defines enzyme's CS-initiation role; Desbuquois-like severe end
31325655	*Biallelic CSGALNACT1-mutations cause a mild skeletal dysplasia*	Second/juvenile patient; homozygous frameshift; "mild and non-progressive"; WES diagnosis
31705726	*CSGALNACT1-CDG: A mild skeletal dysplasia with advanced bone age*	Two more patients; names CSGALNACT1-CDG; altered CS/DS/HS in fibroblasts; advanced bone age, facial dysmorphism, language delay
21148564	*CSGalNAcT1 necessary for normal endochondral ossification and aggrecan metabolism*	KO mouse: slight dwarfism, growth-plate disorganization, aggrecan changes
20812917	*CSGalNAcT-1 required for normal cartilage development*	KO mouse: ~50% CS reduction; disarranged type-II collagen
28987564	*Roles of CSGalNAcT1 in neuronal regeneration and plasticity*	PNN/plasticity branch; basis for inferred mild neurodevelopmental feature
42094029	*Genetic Skeletal Disorders with Defects in GAG Biosynthesis*	Nosology; linkeropathy classification; differential diagnosis framework
31196143	*Clinical/mutational spectrum of B3GAT3 linkeropathy*	Differential diagnosis (more severe linkeropathy phenotype)
32839819	*CSGalNAcT1 KO milder EAE*	CS neuroimmune biology (model context)
32608461	*Synthesis of linkage-region oligosaccharides*	In vitro substrate work on CS chain initiation

Cancer-focused CSGALNACT1 papers (PMIDs 27655130, 25923053, 25445787) and the tau/Alzheimer study (P 41996943) describe the gene's broader CS biology but are **not** directly relevant to SDJLABA pathology; they are noted for completeness of the gene's functional landscape.

Limitations and Knowledge Gaps

- **Tiny cohort.** Only ~5 patients across 3 reports; prevalence, penetrance, expressivity, and genotype–phenotype correlations are statistically underpowered.
- **No formal nosology codes.** No dedicated Orphanet/ICD/MeSH entry identified; classification relies on OMIM/MONDO.
- **Inferred links.** The mechanistic bases for *advanced bone age* and *language delay* are plausible but not experimentally demonstrated in patients.
- **No QoL / natural-history data.** No validated outcome instruments applied; long-term adult outcomes unknown.
- **No therapeutics.** No treatment has been trialed; management is extrapolated from phenotype.
- **Model gaps.** Mouse KO underrepresents advanced bone age and dysmorphism; no humanized or patient-iPSC chondrocyte/organoid models reported.

Proposed Follow-up Experiments / Actions

1. **International case collection** (GeneMatcher / Matchmaker Exchange) to expand the cohort, quantify penetrance/expressivity, and establish allele-specific genotype–phenotype correlations.
2. **Patient-derived iPSC → chondrocyte/organoid models** to directly test CS-initiation deficiency, growth-plate ECM assembly, and candidate corrective approaches in human cells.
3. **Quantitative GAG/disaccharide profiling** (LC-MS) across multiple patient fibroblast lines and, where possible, cartilage, to standardize a biochemical diagnostic signature (CS/DS/HS ratios).
4. **Skeletal-imaging natural-history study** to characterize advanced bone-age trajectory, final adult height, and orthopedic complication rates.
5. **Mechanistic dissection of the bone-age and neurodevelopmental branches** — e.g., bone-age assessment and behavioral/plasticity phenotyping in conditional Csgalnact1 KO mice (chondrocyte- vs. neuron-specific).

6. **Variant functional calibration** — systematic enzyme-activity assays for all reported and future variants to support ACMG classification and residual-activity-based prognostication.
7. **ClinVar/OMIM/MONDO curation update** and submission of a GeneReviews-style synthesis to consolidate diagnostic and management guidance for this ultra-rare disorder.

Report compiled from 9 confirmed findings and 14 reviewed papers across 5 investigation iterations. Evidence types: human clinical case reports (primary), model-organism (mouse KO), in vitro enzymology/fibroblast, and computational/population-constraint (gnomAD).
